# HCIA-AI V4.0 Certification Practice Exam • Chapter 5

Module 5: AI Business Process & Prompt Engineering (100 Questions & Answer Key)

Curated by: Umed Jamal Nouri • v1.1 (Nov 2026) • github.com/UmedJaMala

## Question 1

TRUE OR FALSE • 16 Points

In Data Parallelism, the model architecture is partitioned layer by layer across devices while the entire dataset remains on a single GPU.

**A.** True
**B.** False

## Question 2

TRUE OR FALSE • 16 Points

In Pipeline Parallelism, the weight parameters within each individual layer are partitioned across multiple GPUs for matrix multiplication.

**A.** True
**B.** False

## Question 3

TRUE OR FALSE • 16 Points

Model fine-tuning primarily adjusts the number of neural network layers and changes the overall network topology rather than tuning existing weight parameters.

**A.** True
**B.** False

## Question 4

TRUE OR FALSE • 16 Points

The core idea of Adapter Tuning is to insert small, trainable adapter sub-modules into fixed layers of a pre-trained model while freezing the original backbone weights.

**A.** True
**B.** False

## Question 5

TRUE OR FALSE • 16 Points

In traditional AI business development, Requirement Analysis is the starting phase where business objectives are translated into clear machine learning problem formulations.

**A.** True
**B.** False

**Page Answers:** **Q1**: False • **Q2**: False • **Q3**: False • **Q4**: True • **Q5**: True

### Question 6

TRUE OR FALSE • 16 Points

Concept Drift refers to the physical hardware failure of hard drives and GPUs during model serving.

**A.** True
**B.** False

### Question 7

TRUE OR FALSE • 16 Points

Tensor Parallelism splits individual weight matrices (such as attention projection matrices or MLP layers) across multiple accelerators to reduce per-device memory usage.

**A.** True
**B.** False

### Question 8

TRUE OR FALSE • 16 Points

Retrieval-Augmented Generation (RAG) completely eliminates the need for an external knowledge base by embedding all world facts directly into model activation weights.

**A.** True
**B.** False

### Question 9

TRUE OR FALSE • 16 Points

In Prompt Engineering, Zero-Shot prompting provides multiple input-output demonstration examples before presenting the actual query.

**A.** True
**B.** False

### Question 10

TRUE OR FALSE • 16 Points

In the AI business development pipeline, manual Feature Engineering (Optional) is strictly mandatory for deep learning Computer Vision (CV) and NLP tasks, rather than for Recommendation Systems.

**A.** True
**B.** False

**Page Answers:** **Q6**: False • **Q7**: True • **Q8**: False • **Q9**: False • **Q10**: False

### Question 11

TRUE OR FALSE • 16 Points

Chain of Thought (CoT) prompting encourages language models to generate explicit intermediate reasoning steps, improving performance on multi-step logic and math problems.

**A.** True
**B.** False

### Question 12

TRUE OR FALSE • 16 Points

Zero-Shot-CoT requires developers to manually author dozens of detailed step-by-step reasoning exemplars for every prompt.

**A.** True
**B.** False

### Question 13

TRUE OR FALSE • 16 Points

Reinforcement Learning from Human Feedback (RLHF) aligns large model outputs with human intent, helpfulness, and safety guidelines.

**A.** True
**B.** False

### Question 14

TRUE OR FALSE • 16 Points

In the RLHF pipeline, the Reward Model (RM) is trained on human pairwise preference rankings comparing different model responses to the same prompt.

**A.** True
**B.** False

### Question 15

TRUE OR FALSE • 16 Points

LoRA (Low-Rank Adaptation) updates 100% of all pre-trained model weights, requiring complete retraining of all attention and feedforward layers.

**A.** True
**B.** False

**Page Answers:** **Q11**: True • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: False

### Question 16

TRUE OR FALSE • 16 Points

A/B testing is an offline mathematical proof technique where models are tested without ever exposing them to real user traffic.

**A.** True
**B.** False

### Question 17

TRUE OR FALSE • 16 Points

Prompt Injection is a security vulnerability where malicious user inputs trick a language model into ignoring its developer system instructions and executing unauthorized actions.

**A.** True
**B.** False

### Question 18

TRUE OR FALSE • 16 Points

Data Contamination refers to physical dust and humidity damaging server hard drives in data centers.

**A.** True
**B.** False

### Question 19

TRUE OR FALSE • 16 Points

Prompt Tuning prepends continuous learnable virtual token embeddings to the input sequence while freezing the underlying language model weights.

**A.** True
**B.** False

### Question 20

TRUE OR FALSE • 16 Points

Prefix Tuning attaches learnable prefix vectors to the keys and values of attention layers in each Transformer block.

**A.** True
**B.** False

**Page Answers:** **Q16**: False • **Q17**: True • **Q18**: False • **Q19**: True • **Q20**: True

### Question 21

TRUE OR FALSE • 16 Points

Supervised Fine-Tuning (SFT) transforms raw pre-trained models into conversational instruction-following assistants using paired (instruction, output) data.

**A.** True
**B.** False

### Question 22

TRUE OR FALSE • 16 Points

Large language models deployed in enterprise production do not require monitoring because foundation models are permanently immune to factual decay.

**A.** True
**B.** False

### Question 23

TRUE OR FALSE • 16 Points

The ReAct framework combines reasoning traces and task-specific external tool actions in an interleaved loop to enable autonomous agent problem-solving.

**A.** True
**B.** False

### Question 24

TRUE OR FALSE • 16 Points

In large model training, 3D Parallelism refers to at the same time combining Data Parallelism, Tensor Parallelism, and Pipeline Parallelism across thousands of accelerators.

**A.** True
**B.** False

### Question 25

TRUE OR FALSE • 16 Points

Evaluating large models using an LLM-as-a-Judge approach is completely immune to self-enhancement bias or position bias.

**A.** True
**B.** False

**Page Answers:** **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: True • **Q25**: False

### Question 26

SINGLE-ANSWER QUESTION • 16 Points

Within the broad landscape of large language models, what is the primary goal of Supervised Fine-Tuning (SFT)?

**A.** Substituting internal attention layers with traditional rule-based scripts
**B.** Compressing the total parameter capacity of the model by ninety percent using pruning
**C.** Adapting the pre-trained model to follow specific user instructions and conversational formats
**D.** Re-training all weights from scratch without pre-trained checkpoints

### Question 27

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements is TRUE regarding parallel training strategies for large models?

**A.** Data parallelism divides model layers across devices, whereas tensor parallelism splits batches
**B.** Tensor parallelism splits single layers, whereas pipeline parallelism partitions sequential layers
**C.** Large models can only be trained on standalone workstation GPUs without distributed clustering
**D.** Pipeline parallelism executes matrix multiplications synchronously on single-core microcontrollers

### Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which distributed training strategy partitions the sequential layers of a deep neural network across multiple accelerators like stages in an assembly line?

**A.** Pipeline Parallelism
**B.** K-Means Clustering
**C.** Data Parallelism
**D.** Zero-Shot Prompting

### Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which PEFT method freezes original weights W_0 and injects trainable rank decomposition matrices B and A such that W = W_0 + B * A (where rank r << min(d, k))?

**A.** Full fine-tuning
**B.** Gradient descent
**C.** LoRA
**D.** Grid search

### Question 30

SINGLE-ANSWER QUESTION • 16 Points

What is the primary advantage of Parameter-Efficient Fine-Tuning (PEFT) compared to Full Fine-Tuning?

**A.** Greatly lowers GPU memory and trainable parameters while preventing catastrophic forgetting
**B.** Completely eliminates the requirement for labeled training datasets through unsupervised clustering
**C.** Enforces simultaneous end-to-end retraining of all backbone layers across distributed cloud clusters
**D.** Scales up total backbone parameter count tenfold by stacking additional multi-head attention layers

**Page Answers:** **Q26**: C • **Q27**: B • **Q28**: A • **Q29**: C • **Q30**: A

### Question 31

SINGLE-ANSWER QUESTION • 16 Points

In the traditional AI project lifecycle, what phase immediately follows Model Training before production deployment?

- **A.** Problem Requirement Analysis
- **B.** Hyperparameter Sensitivity Profiling
- **C.** Model Evaluation and Validation
- **D.** Data Collection

### Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which prompting technique enables multi-step deductive reasoning by simply appending 'Let us think step by step'?

- **A.** Post-Training Vector Quantization
- **B.** Zero-Shot Chain-of-Thought
- **C.** In-Context Few-Shot Feature Alignment
- **D.** Differential Data Anonymization

### Question 33

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of Retrieval-Augmented Generation (RAG) in enterprise AI applications?

- **A.** Rewriting native operating system kernel scheduling algorithms using interpreted Python code
- **B.** Querying external knowledge stores to inject real-time, domain context and reduce hallucination
- **C.** Deleting outdated customer database records automatically based on temporal retention rules
- **D.** Training foundational transformer base models from scratch using random initialization weights

### Question 34

SINGLE-ANSWER QUESTION • 16 Points

In the RLHF (Reinforcement Learning from Human Feedback) workflow, which algorithm optimizes the policy model?

- **A.** Ordinary Least Squares Linear Regression
- **B.** K-Means Centroid Coordinate Clustering
- **C.** Information Gain Ratio Decision Tree
- **D.** Proximal Policy Optimization

### Question 35

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Reward Model' in the RLHF process?

- **A.** A model trained on human preference rankings to output a scalar quality score for responses
- **B.** An ACID relational transactional accounting database
- **C.** A server billing script calculating enterprise electricity consumption during training runs
- **D.** A hyperparameter grid search module tuning learning rates across validation folds

**Page Answers:** **Q31**: C • **Q32**: B • **Q33**: B • **Q34**: D • **Q35**: A

### Question 36

SINGLE-ANSWER QUESTION • 16 Points

What security threat occurs when an adversary crafts adversarial prompts to bypass safety guardrails and elicit restricted or dangerous content?

- **A.** Batch Normalization
- **B.** Cross-Validation
- **C.** Jailbreaking
- **D.** Weight Decay

### Question 37

SINGLE-ANSWER QUESTION • 16 Points

What are the four core structural components of a well-designed professional prompt?

- **A.** HTML, CSS, JavaScript, and PHP
- **B.** IP Address, Port Number, Gateway, and Subnet Mask
- **C.** Instruction, Context, Input Data, and Output Indicator
- **D.** Loss Function, Optimizer, Batch Size, and Epoch Count

### Question 38

SINGLE-ANSWER QUESTION • 16 Points

What is the primary operational mechanism of Adapter Tuning?

- **A.** Rebuilding the entire tokenizer vocabulary dictionary by parsing multi-language web archives
- **B.** Inserting small bottleneck feedforward layers into the transformer while keeping backbone weights frozen
- **C.** Pruning all attention heads from the transformer to force pure sequential recurrent processing
- **D.** Multiplying base model learning rates by one thousand to accelerate gradient convergence speed

### Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an advantage of Zero-Shot-CoT compared to Few-Shot-CoT?

- **A.** It offers better task scalability and flexibility
- **B.** It always guarantees strictly superior accuracy over hand-crafted Few-Shot-CoT in every benchmark
- **C.** It requires no manual engineering of multi-step problem exemplars
- **D.** It is simple to implement across diverse tasks using a standardized prompt suffix

### Question 40

SINGLE-ANSWER QUESTION • 16 Points

What type of fine-tuning updates all model parameters at the same time across the entire network architecture?

- **A.** Low-Rank Adaptation Fine-Tuning
- **B.** Prefix Tuning Parameter Adaptation
- **C.** Full Model Fine-Tuning
- **D.** Prompt Tuning Soft Embedding Strategy

**Page Answers:** **Q36**: C • **Q37**: C • **Q38**: B • **Q39**: B • **Q40**: C

### Question 41

SINGLE-ANSWER QUESTION • 16 Points

In enterprise AI production monitoring, what does 'Data Drift' refer to?

- **A.** Physical signal attenuation occurring along undersea fiber-optic telecommunication cables
- **B.** Changes in input feature distribution P(X) over time compared to the original training distribution
- **C.** Floating-point numerical precision rounding errors across different CPU registers
- **D.** Accidental deletion of validation split partitions from cloud object storage repositories

### Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which evaluation approach uses state-of-the-art foundation models (such as GPT-4) to grade and evaluate responses generated by candidate models?

- **A.** Keyword matching
- **B.** Grid search
- **C.** LLM-as-a-Judge
- **D.** Unit testing

### Question 43

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of the 'Instruction' component in a prompt?

- **A.** Supplying pseudorandom seed integers to initialize GPU CUDA calculation kernels
- **B.** Specifying the exact task, operational behavior, or objective the model is required to perform
- **C.** Allocating static memory blocks within dedicated hardware video RAM buffers
- **D.** Defining firewall subnet routing policies for outbound API transmission endpoints

### Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which vector database operation is used in RAG to find the most relevant context documents?

- **A.** Dense vector semantic similarity search
- **B.** Exact lexical keyword matching using inverted boolean indices
- **C.** Exact regular expression wildcard string matching
- **D.** Uniform random sampling across database records

### Question 45

SINGLE-ANSWER QUESTION • 16 Points

Why is 'Human-in-the-Loop' (HITL) crucial in high-risk AI business processes?

- **A.** To provide human expert oversight, safety validation, and ethical accountability for critical model decisions
- **B.** To automatically retrain neural network weights in real time without human evaluation
- **C.** To execute scalar arithmetic calculations on host CPU registers
- **D.** To guarantee 100% rule-based elimination of all software bugs in database schemas

**Page Answers:** **Q41**: B • **Q42**: C • **Q43**: B • **Q44**: A • **Q45**: A

## Question 46

SINGLE-ANSWER QUESTION • 16 Points

What is the primary goal of the Pre-training stage in foundation model development?

- **A.** Learning universal language representations and world knowledge from vast unlabeled datasets
- **B.** Fitting an isolated decision tree classifier to predict categorical churn on tabular datasets
- **C.** Restricting model output syntax strictly to RFC-compliant JSON dictionary structures
- **D.** Memorizing fixed polite greeting phrases tailored for enterprise customer service chatbots

## Question 47

SINGLE-ANSWER QUESTION • 16 Points

In LoRA, what is the hyperparameter 'rank' (r)?

- **A.** The inner low dimension of decomposition matrices A and B
- **B.** The total count of physical GPU acceleration nodes allocated in the cluster rack
- **C.** The relative performance ranking score achieved by the model on public benchmarks
- **D.** The batch size threshold configured for distributed asynchronous gradient updates

## Question 48

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes 'Direct Preference Optimization' (DPO)?

- **A.** A sorting algorithm optimizing B-tree indices in relational database management systems
- **B.** An analytical closed-form solver calculating regression weights via ordinary least squares
- **C.** A graph search algorithm finding minimum spanning trees across distributed networks
- **D.** An alignment algorithm directly optimizing policy weights on human preferences without a separate reward model

## Question 49

SINGLE-ANSWER QUESTION • 16 Points

What does MMLU (Massive Multitask Language Understanding) evaluate in large language models?

- **A.** Cross-domain knowledge and problem-solving across 57 academic subjects
- **B.** Network packet routing hop counts between routers
- **C.** Graphics processing unit rendering framerates and polygon rasterization speeds in video games
- **D.** Input/output operations per second across high-throughput solid-state storage arrays

## Question 50

SINGLE-ANSWER QUESTION • 16 Points

In a RAG pipeline, what component is responsible for transforming raw text chunks into dense vector embeddings?

- **A.** Reranker
- **B.** Tokenizer
- **C.** Vector database
- **D.** Embedding model

**Page Answers:** **Q46**: A • **Q47**: A • **Q48**: D • **Q49**: A • **Q50**: D

### Question 51

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an example of an Output Indicator in prompt engineering?

- **A.** 'Translate the following business email into fluent Spanish.'
- **B.** 'Provide the response in valid JSON format containing keys: summary and action_items.'
- **C.** 'You are a certified senior cloud architect with twenty years of experience.'
- **D.** 'Input context: The quick brown fox jumps over the lazy dog in the meadow.'

### Question 52

SINGLE-ANSWER QUESTION • 16 Points

What is the primary risk of 'Over-Refusal' in safety-aligned foundation models?

- **A.** The model exhaustively consumes all available host memory during batch token generation
- **B.** The model emits vulgar phrases and unmoderated offensive tokens to end users
- **C.** The model triggers unhandled kernel panics across underlying distributed server runtimes
- **D.** The model rejects benign, completely safe queries due to overly sensitive safety guardrails

### Question 53

SINGLE-ANSWER QUESTION • 16 Points

In distributed large model training, which collective communication operation gathers and sums gradients across all worker nodes in Data Parallelism?

- **A.** Broadcast
- **B.** Scatter
- **C.** Gather
- **D.** AllReduce

### Question 54

SINGLE-ANSWER QUESTION • 16 Points

What is 'Catastrophic Forgetting' in neural networks and foundation models?

- **A.** Loss of saved model checkpoint files caused by unrecoverable physical disk sectors
- **B.** Omission of standard library import statements in application deployment scripts
- **C.** System authentication failure resulting from forgotten user database passwords
- **D.** The tendency of a model to lose previously learned capabilities upon learning new tasks

### Question 55

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a classic Python framework for building LLM and RAG application workflows with composable chains and agents?

- **A.** LangChain Framework
- **B.** Matplotlib Library
- **C.** LightGBM Framework
- **D.** Scikit-Learn Toolkit

**Page Answers:** **Q51**: B • **Q52**: D • **Q53**: D • **Q54**: D • **Q55**: A

### Question 56

SINGLE-ANSWER QUESTION • 16 Points

What is the primary metric for evaluating code generation capabilities in models like DeepSeek-Coder?

- **A.** Perplexity
- **B.** Mean squared error
- **C.** Pass@k
- **D.** BLEU score

### Question 57

SINGLE-ANSWER QUESTION • 16 Points

In traditional machine learning project deployment, what is 'Canary Deployment'?

- **A.** Replacing software inference models with fixed-function application-specific integrated circuits
- **B.** Terminating all active web production servers at the same time during peak commercial traffic hours
- **C.** Replacing the entire production cluster at the same time without staged traffic routing
- **D.** Directing a small percentage of live traffic to the new model to verify stability before full rollout

### Question 58

SINGLE-ANSWER QUESTION • 16 Points

Which prompt engineering technique prompts the model to assign itself a specific expert identity (e.g., 'Act as an expert cybersecurity auditor')?

- **A.** Token Pruning
- **B.** Data Imputation
- **C.** Temperature 0 Sampling
- **D.** Role Prompting

### Question 59

SINGLE-ANSWER QUESTION • 16 Points

What is the impact of setting LLM inference 'Temperature' to 0?

- **A.** The model generates highly divergent, chaotic text with maximum Shannon entropy values
- **B.** Token generation velocity drops to zero because computation threads enter permanent sleep
- **C.** The model produces deterministic, greedy responses by always selecting the highest-probability token
- **D.** Model checkpoint weights fail cryptographic hash integrity checks

### Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

In Large Language Model text generation, which decoding hyperparameters directly regulate the diversity and randomness of sampled tokens? (Multiple Choice)

- **A.** Temperature
- **B.** Top-P or Nucleus Sampling
- **C.** Top-K Sampling
- **D.** Weight Decay Coefficient

**Page Answers:** **Q56**: C • **Q57**: D • **Q58**: D • **Q59**: C • **Q60**: ABC

### Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are dedicated vector database technologies commonly used for high-dimensional semantic similarity search in RAG?

- **A.** Milvus
- **B.** SQLite
- **C.** Chroma
- **D.** MySQL

### Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following data sources are commonly aggregated and curated for Large Language Model self-supervised pre-training?

- **A.** Filtered, deduplicated, and toxicity-cleansed web text crawl datasets
- **B.** Curated open-source code repositories, technical documentation, and commit histories
- **C.** Encrypted private cryptographic SSL handshake session payloads
- **D.** High-quality digitized books, peer-reviewed academic papers, and encyclopedic reference collections

### Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In Large Language Model architecture and application design, why is the Context Window size a critical technical specification?

- **A.** It establishes the maximum combined token capacity for input prompts, retrieved external context, and generated responses
- **B.** Standard full self-attention memory and compute complexity scales quadratically with respect to context sequence length
- **C.** It directly bounds the volume of long-document grounding context that can be injected in RAG workflows
- **D.** It fixes the physical silicon substrate surface area and transistor die packaging thickness

### Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In an enterprise AI lifecycle and MLOps architecture, which components manage system monitoring and production reliability?

- **A.** Real-time logging of inference latency, token throughput, and cost consumption
- **B.** Periodic physical power-cycling of isolated database compute nodes
- **C.** Data distribution drift detection and model output degradation monitoring
- **D.** Automated fallback routing and circuit-breaker mechanisms for high-availability serving

### Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

In modern Mixture of Experts (MoE) architectures (such as DeepSeekMoE), which benefits are achieved by fine-grained expert segmentation?

- **A.** Increased specialized expert diversity and higher representation capacity per parameter budget
- **B.** Finer granularity in routing decisions, enabling more precise factual and linguistic specialization
- **C.** Complete elimination of multi-head attention mechanisms from the transformer backbone
- **D.** Restricting inference execution strictly to single-threaded CPU instruction sets

**Page Answers:** **Q61**: AC • **Q62**: ABD • **Q63**: ABC • **Q64**: ACD • **Q65**: AB

### Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about large foundation models are true?

- **A.** They possess significantly larger parameter counts than traditional deep learning models
- **B.** They eliminate the need for token embeddings and positional encodings
- **C.** They exhibit emergent capabilities at massive scale
- **D.** They demonstrate strong cross-task generalization through pre-training

### Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following domains can foundation model technologies be applied to?

- **A.** Multimodal creative content creation
- **B.** Intelligent customer service and conversational assistants
- **C.** Automated software programming and code generation
- **D.** Scientific literature analysis and research hypothesis generation

### Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements accurately characterize Parameter-Efficient Fine-Tuning (PEFT) methods covered in HCIA-AI V4.0?

- **A.** Adapter Tuning inserts small trainable modules into pre-trained model layers while freezing the main backbone
- **B.** Prefix Tuning and Prompt Tuning avoid altering internal Transformer block structure by adding trainable virtual/prompt tokens
- **C.** In LoRA, matrix A starts with random Gaussian values and matrix B starts as a zero matrix before merging BA with frozen weights
- **D.** Full-parameter fine-tuning updates strictly fewer weights than Prompt Tuning while freezing backbone layers

### Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following Parameter-Efficient Fine-Tuning (PEFT) methods freeze the base model weights and introduce trainable low-rank or prompt matrices?

- **A.** LoRA
- **B.** Full-Parameter End-to-End Retraining
- **C.** Prompt Tuning
- **D.** Random Gaussian Weight Re-initialization

### Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

The development and enterprise operation of large models requires which of the following lifecycle stages?

- **A.** Pre-training, instruction fine-tuning, and alignment
- **B.** Model serving, monitoring, and prompt engineering
- **C.** Data collection, cleaning, and filtering
- **D.** Safety testing, model evaluation, and monitoring

**Page Answers:** **Q66**: ACD • **Q67**: ABCD • **Q68**: ABC • **Q69**: AC • **Q70**: ABCD

### Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following distributed parallelism strategies partition model layers across pipeline stages and tensor operations across multiple accelerators respectively?

- **A.** Unconstrained Greedy Decoding
- **B.** Data Anonymization and Token Masking
- **C.** Pipeline Parallelism
- **D.** Tensor Parallelism

### Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

In the standard RLHF (Reinforcement Learning from Human Feedback) process, which training stages follow Supervised Fine-Tuning (SFT)?

- **A.** Raw Web Data Scraping and Filtering
- **B.** Reward Model Training on human preference comparisons
- **C.** Static C++ Kernel Compilation
- **D.** Policy Optimization using PPO

### Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

A standard enterprise RAG (Retrieval-Augmented Generation) pipeline mainly consists of which core functional modules?

- **A.** Pre-training module
- **B.** Retrieval module
- **C.** Generation module
- **D.** Fine-tuning module

### Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are advantages of Retrieval-Augmented Generation (RAG) over full model fine-tuning?

- **A.** Enables verifiable source attribution and citations in answers
- **B.** Greatly lowers compute costs compared to continually retraining large models
- **C.** Allows updating knowledge in real time by simply adding documents to the vector database
- **D.** Eliminates the need for a generative language model completely

### Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key dimensions evaluated when assessing large language models?

- **A.** Helpfulness
- **B.** Honesty
- **C.** Efficiency
- **D.** Harmlessness

**Page Answers:** **Q71**: CD • **Q72**: BD • **Q73**: BC • **Q74**: ABC • **Q75**: ABD

### Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common Prompt Engineering techniques?

- **A.** Role-Based Prompting
- **B.** Few-Shot In-Context Learning
- **C.** Zero-Shot Prompting
- **D.** Chain-of-Thought Prompting

### Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent common forms of Prompt Injection security threats in LLM applications?

- **A.** Direct Prompt Injection
- **B.** Indirect Prompt Injection
- **C.** Database Schema Migration Lockout
- **D.** Ethernet Network Cable Disconnection

### Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Zero-Shot-CoT are true?

- **A.** It has high generalization across diverse reasoning tasks
- **B.** It requires no manually authored input-output reasoning exemplars
- **C.** It triggers step-by-step reasoning using prompt phrases like 'Let's think step by step'
- **D.** It completely eliminates all reasoning capability

### Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks are performed during the Data Preparation phase of a large language model project?

- **A.** Text extraction from diverse file formats
- **B.** Quality filtering and removal of toxic or low-quality text
- **C.** Tokenization using algorithms like BPE
- **D.** Deduplication of identical and near-duplicate documents

### Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common challenges encountered in Large Model Evaluation?

- **A.** High cost and subjectivity of human expert evaluation
- **B.** Inability to count the number of test questions
- **C.** Benchmark saturation on popular public test sets
- **D.** Test set contamination in pre-training datasets

**Page Answers:** **Q76**: ABCD • **Q77**: AB • **Q78**: ABC • **Q79**: ABCD • **Q80**: ACD

## Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are defining mathematical properties of LoRA (Low-Rank Adaptation)?

- **A.** Multiplies total trainable parameters tenfold by adding layers
- **B.** Freezes the original pre-trained weight matrix W_0
- **C.** Decomposes weight updates delta W into the product of two low-rank matrices B * A
- **D.** Requires training all backbone parameters from epoch zero

## Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key features of the ReAct (Reason + Act) prompting framework?

- **A.** Observing tool outputs to update the reasoning trajectory
- **B.** Disabling all model attention layers
- **C.** Executing external tools and APIs to retrieve real-time data
- **D.** Generating reasoning thoughts before selecting actions

## Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common online deployment strategies for safely rolling out new AI model versions to production users?

- **A.** Direct overwriting
- **B.** Manual cutover
- **C.** Canary deployment
- **D.** Blue-green deployment

## Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common evaluation benchmarks for large language models?

- **A.** HumanEval
- **B.** Massive Multitask Language Understanding
- **C.** Cinebench
- **D.** Grade School Math 8K

## Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following techniques can reduce prompt injection attacks?

- **A.** Directly pasting raw user input as system instructions
- **B.** Output content filtering and safety checks
- **C.** Using separate system instruction channels that cannot be overridden by user inputs
- **D.** Input validation and prompt delimiters

**Page Answers:** **Q81**: BC • **Q82**: ACD • **Q83**: CD • **Q84**: ABD • **Q85**: BCD

### Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following automated n-gram overlap metrics are commonly used to evaluate machine translation and text summarization quality?

- **A.** Bilingual Evaluation Understudy
- **B.** Recall-Oriented Understudy for Gisting Evaluation
- **C.** Silhouette Cluster Separation Index
- **D.** K-Means Elbow Inertia

### Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Vector Databases in RAG systems are true?

- **A.** They enable fast semantic retrieval over millions of enterprise documents
- **B.** They support Approximate Nearest Neighbor search algorithms
- **C.** They index high-dimensional embeddings generated by neural networks
- **D.** They can only store 10-word plain text files

### Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors should be considered during model selection in an enterprise AI project?

- **A.** Target business accuracy and latency requirements
- **B.** Open-source licensing and commercial compliance
- **C.** Favorite color of the software engineer
- **D.** Available computational budget for training and inference

### Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary operational advantages of Adapter Tuning?

- **A.** Keeps the base model backbone frozen, preventing catastrophic forgetting
- **B.** Multiplies total model parameters by ten to maximize capacity
- **C.** Guarantees zero floating-point computation latency
- **D.** Significantly reduces storage and training compute requirements

### Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common causes of Concept Drift in production machine learning systems?

- **A.** Changes in consumer shopping behavior over holiday seasons
- **B.** Evolution of user vocabulary and slang on social media
- **C.** Macroeconomic fluctuations and policy changes
- **D.** The screen resolution of client laptops

**Page Answers:** **Q86**: AB • **Q87**: ABC • **Q88**: ABD • **Q89**: AD • **Q90**: ABC

### Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent core sequential training stages in the development of modern Large Language Models?

- **A.** Pre-training
- **B.** Supervised Fine-Tuning
- **C.** Data deduplication
- **D.** RLHF alignment

### Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methodologies are established techniques for aligning Large Language Models with human values and helpfulness?

- **A.** Reinforcement Learning from Human Feedback
- **B.** Direct Preference Optimization
- **C.** Reinforcement Learning from AI Feedback
- **D.** Rejection Sampling and Best-of-N Policy Optimization

### Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

The parameter-efficient fine-tuning technique that decomposes weight updates into two low-rank matrices is called ____.

**Answer:** _____

### Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

The enterprise technique that retrieves relevant factual documents from an external knowledge base to ground model generation is called ____ (Retrieval-Augmented Generation).

**Answer:** _____

### Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In distributed model training, when a single layer's matrix operations are split across multiple GPUs, this is called ____ Parallelism.

**Answer:** _____

**Page Answers:** **Q91**: ABD • **Q92**: ABCD • **Q93**: LoRA (or Low-Rank Adaptation) • **Q94**: RAG • **Q95**: Tensor

### Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In distributed model training, when sequential layers of a neural network are partitioned across sequential devices like an assembly line, this is called ____ Parallelism.

**Answer:** _____

### Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

Implemented in C/C++, the standalone deployment tool ____ runs Large Language Model inference locally or in the cloud without requiring an AI development framework.

**Answer:** _____

### Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

The security vulnerability where an attacker uses crafted prompts to bypass guardrails and violate system rules is called ____ (or Prompt Injection).

**Answer:** _____

### Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In prompt engineering, the strategy of providing a few solved demonstration examples inside the prompt is known as ____-Shot Prompting.

**Answer:** _____

### Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In the traditional AI business process, after model training and offline validation, models are often evaluated on live traffic using ____ testing.

**Answer:** _____

**Page Answers:** **Q96**: Pipeline • **Q97**: llama.cpp • **Q98**: Jailbreaking (or Jailbreak) • **Q99**: Few • **Q100**: A/B